

IS 302-2-45 : 1994

भारतीय मानक

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घरेलू और समान विद्युत साधनों की सुरक्षा

भाग 2 विशेष अपेक्षाएँ

अनुभाग 45 बिजली के तापन औज़ार

Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 45 Electric Heating Tools

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BUREAU OF INDIAN STANDARDS
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Price Group 5

Electrical Appliances Sectional Committee, ET 32

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrotechnical Division Council. This standard specifies general and safety requirements. The composite standard specifying the requirement of general, safety and performance characteristics of electric soldering irons are covered under IS 959 : 1994. The composite standards covering general safety and performance requirements of other heating tools covered under the scope of this standard will be brought out at a later date after establishing the performance characteristics and their requirements.

This standard is to be read in conjunction with IS 302-1 (1979) 'Safety of household and similar electrical appliances : Part 1 General requirements'. For the sake of convenience, the clauses of this standard correspond to those of IS 302-1 (1979) instead of reproducing full text of each clause, clauses of 302-1 (1979) which are applicable (which means that relevant provisions of the clause apply) or not applicable and the sub-clause or portions thereof which are not applicable are indicated as under:

- a) In the case of a clause where it is applicable or not applicable, the wording used is 'This clause of IS 302-1 (1979) is applicable/not applicable'.
- b) In case of a sub-clause or part thereof 'not applicable' wherever a sub-clause of IS 302-1 (1979) is to be replaced by a new text, it has been indicated as under:

'Replacement' — followed by the new text.

Any addition to the existing provisions of a sub-clause of IS 302-1 (1979) has been indicated as under:

'Addition' — followed by the text of the additional matter.

Clauses/tables which are additional to those of IS 302-1 (1979) are numbered starting from 101 and additional sub-clauses are numbered with main clause number followed by 101, 102, etc, for example 7.101.

While preparing this standard assistance has been derived from IEC Pub 335-2-45 (1986) 'Safety of household and similar electrical appliances : Part 2 Particular requirements for electric heating tools' and IEC Pub 335-1 (1976) 'Safety of household and similar electrical appliances : Part 1 General Requirements' issued by the International Electrotechnical Commission.

The Indian Standards which are necessary adjuncts to this standard are given in Annex AA.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of the test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in the standard.

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Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 45 Electric Heating Tools

1 SCOPE

This clause of IS 302-1 (1979) is applicable except as follows :

1.1 Replacement

This standard applies to portable electric heating tools and similar appliances, with or without motors, their rated voltage not exceeding 250 V ac single phase or dc.

Portable electric heating tools and similar appliances, hereinafter referred to as tools, which can be mounted on a support for use as fixed tools without any alteration of the tool itself, are within the scope of this standard.

Tools not intended for normal household use, but which nevertheless may be a source of danger to the public, such as tools intended to be used by laymen in shops, in light industry and on farms, are also within the scope of this standard.

Examples of tools that are within the scope of this standard are:

- Soldering irons;
- Desoldering irons;
- Soldering guns;
- Conduit-soldering tools;
- Thermoplastic conduit-welding tools;
- Film-welding tools;
- Household film-welding appliances;
- Plastic-cutting tools;
- Branding tools;
- Burning-in pens;
- Stripping pliers;
- Paint strippers;
- Fire lighters;
- Heat guns;
- Glue guns; and
- Dehorning tools.

This standard does not take into account the special hazards which exist in nurseries and other places where there are young children or aged or infirm persons without supervision; in such cases, additional requirements may be necessary.

This standard does not apply to :

- Tools for high-frequency heating;
- Arc welding and carbon-arc brazing equipment; and

- Tools intended to be used in explosive atmospheres.

For tools intended to be used in vehicles or on board ships or aircraft, additional requirements may be necessary.

2 TERMINOLOGY

This clause of IS 302-1-1979 is applicable except as follows :

2.29 Replacement

Conditions of adequate heat discharge denote the conditions that apply when the tool is operated in still air under the following conditions :

Tools delivered together with a stand, either incorporated in the tool or separate, are operated on the stand, unless otherwise specified.

Tools not delivered together with a stand and tools designed to be placed on a workbench are not operated on a stand.

Other tools are operated in accordance with the manufacturer's instructions, unless otherwise specified.

Soldering irons and similar hand-held tools provided with a heated tip, other than tools provided with a biased-off switch, are operated until steady conditions are established.

Soldering guns and plastic-cutting tools provided with a biased-off switch, are operated intermittently according to the rated 'ON' and 'OFF' periods, but with a minimum of 12 s on and a maximum of 48 s off, until steady conditions are established, except that for soldering guns the temperature at the tip is at least 300°C at the end of the first 'ON' period.

Household film-welding appliances are operated intermittently according to the rated 'ON' and 'OFF' periods, the 'ON' period being at least equal to the time necessary for welding together two sheets of high-pressure polyethylene, each sheet having a thickness of 50 µm, and the 'OFF' period being not longer than 2 minutes, until steady conditions are established.

The 'ON' period includes the time necessary for the operation of any suction device, in accordance with the manufacturer's instructions. Heat guns are suspended in still air so that the hot air flow is horizontal.

Fire lighters are operated to ignite charcoal which is placed in a tray having a diameter of 600 mm and a side 50 mm high. The pile of charcoal is 500 mm in diameter and 100 mm high at its centre.

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- Contact fire lighters are arranged horizontally with the heating element centrally positioned in the charcoal. They are operated for 30 minutes.
- Hot-air fire lighters are arranged horizontally or with the hot air flow directed downwards to the maximum permitted by the construction, but not more than 45°, whichever is the more unfavourable. They are operated for 10 minutes.

Add

Other tools are operated until steady conditions are established, irrespective of any marking.

Additional sub-clauses

2.101 Soldering iron denotes a tool, the soldering tip of which is heated by a heating element.

2.102 Desoldering iron denotes a soldering iron provided with a device for removing molten solder.

2.103 Soldering gun denotes a tool incorporating a transformer, the secondary circuit of which comprises the soldering tip.

2.104 Conduit-soldering tool denotes a tool intended to be used for joining metal piping by means of solder.

2.105 Film-welding tool denotes a tool intended to be used for welding or cutting thin thermoplastic sheets by means of electrically heated parts between which the material is clamped.

2.106 Household film-welding appliance denotes an appliance for household use only and intended to be used for welding or cutting thin thermoplastic sheets by means of electrically heated parts between which the material is clamped.

2.107 Branding tool denotes a tool intended to be used for marking wood, plastics, leather and the like by means of a metal part heated by a heating element.

2.108 Burning-in pen denotes a tool intended to be used for marking wood, leather and the like by means of a heated tip.

2.109 Stripping pliers denotes a tool intended to be used for stripping insulation from a conductor by means of heated jaws.

2.110 Dehorning tool denotes a tool intended to be used for burning out horn buds.

2.111 Biased-off switch denotes a switch which automatically returns to the 'off' position as soon as its actuating number is released.

2.112 Heat gun denotes a tool that produces a jet of hot air.

NOTE - Examples of use are for fusing materials and for softening paint or plastics.

2.113 Paint stripper denotes an electrically heated tool for softening paint.

NOTE - Paint strippers may incorporate a scraper.

2.114 Fire lighter denotes an electrically heated tool for igniting solid fuels such as charcoal or wood.

2.115 Contact fire lighter denotes a fire lighter incorporating a heating element which is in direct contact with the fuel.

2.116 Hot-air fire lighter denotes a fire lighter incorporating a fan and a heating element, which blows hot air at the fuel.

2.117 Thermoplastic conduit-welding tool denotes a tool where the welding process is made by partly melting the thermoplastic material of a separate fitting.

The fitting has an integral heating wire and is used only once.

3 GENERAL REQUIREMENT

This clause of IS 302-2-1 (1979) is applicable.

4 GENERAL CONDITIONS FOR THE TESTS

This clause of IS 302-2-1 (1979) is applicable except as follows :

4.2 Addition

An additional sample of a contact fire lighter is required for the test of **21.102**.

5 RATING

5.1 This clause of IS 302-2-1 (1979) is applicable.

6 CLASSIFICATION

This clause of IS 302-2-1 (1979) is applicable.

7 MARKING

This clause of IS 302-2-(1979) is applicable except as follows:

7.1 Addition

Contact fire lighters shall be marked with the maximum length of insertion into the fuel.

Contact fire lighters which are not at least splash-proof shall be marked with :

"DO NOT EXPOSE TO RAIN OR MOISTURE"

7.3 Addition

Thermoplastic conduit-welding tools shall be marked with the types of fittings with which they are to be used and with the corresponding settings.

Each fitting shall be marked with the type of tool with which it is to be used and with its own type reference.

7.5 Addition

The requirement that the input of the appliance in cold condition shall be marked on the appliance when it differs by more than 25 percent from the input at operating temperature does not apply to tools supplied from an incorporated transformer.

7.12 Addition

Tools delivered with a separate stand and not provided with a biased-off switch, shall be accompanied by an instruction sheet including the substance of the following warning :

THIS TOOL MUST BE PLACED ON ITS STAND WHEN NOT IN USE.

Class III dehorning tools shall be accompanied by an instruction sheet including the substance of the following warning :

Any transformer used for the power supply must be a safety isolating transformer having a protection against moisture of at least splash proof construction when the use of a tool may give rise to fire or the creation of

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explosive or toxic gases and other products of combustion, the instruction sheet shall include an appropriate warning that such hazard may occur.

NOTE – This applies to tools used with some plastics, paints and similar substances.

Heat guns shall be provided with an instruction sheet which includes the substance of the following :

- A fire may arise if the heat gun is not used with care;
- Be careful when using the heat gun in places where there are combustible material or where there may be explosive gases. Do not apply to the same place for a long time;
- Heat may be conducted to combustible materials which are out of sight;
- Place the heat gun on its stand after use and allow it to cool down before storing it; and
- Do not leave the heat gun when it is operating.

Fire lighters shall be provided with an instruction sheet which includes the substance of the following :

- Ensure that the fire lighter is properly positioned before energizing;
- Switch off or disconnect before removal from the fire;
- Allow the fire lighter to cool down before storing it; and
- Do not allow the flexible cord or other flammable materials to come into contact with the heated parts.

Thermoplastic conduit welding tools shall be provided with an instruction sheet which includes the substance of the following :

- A welding operation must not be repeated on a fitting because this can result in live parts becoming accessible.

Additional Sub-clauses

7.101 Soldering irons shall be supplied with adequate instructions for safe and proper use.

7.102 Class III soldering irons the user instructions shall include the following warning which shall be given adequate prominence :

‘WARNING’

THIS SOLDERING IRON IF TO BE SUPPLIED FROM THE MAINS ELECTRICITY SUPPLY MUST BE USED ONLY WITH A SAFETY ISOLATING TRANSFORMER COMPLYING WITH IS 1416 : 1993.

Compliance shall be checked by inspection.

7.103 The heating tools may also be marked with the Standard Mark.

7.103.1 The use of the Standard Mark is governed by the provisions of *Bureau of Indian Standards Act, 1986* and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

8 PROTECTION AGAINST ELECTRIC SHOCK

This clause of IS 302-2-1 (1979) is applicable.

9 STARTING OF MOTOR-OPERATED APPLIANCES

This clause of IS 302-1 (1979) is applicable to those appliances incorporating a motor.

10 INPUT AND CURRENT

This clause of IS 302-1 (1979) is applicable.

11 TEMPERATURE RISE

This clause of IS 302-1 (1979) is applicable except as follows :

11.2 Addition

Tools are placed on the floor of the test corner, away from the walls.

11.4 Addition

Soldering guns and other tools supplied from a transformer are operated at a voltage agreed to 1.06 times the rated voltage.

11.8 Replacement :

Instead of the value specified in the table, for pure mica and tightly sintered ceramic material when used as supplementary insulation or reinforced insulation for heating elements, a temperature rise of 600°C is allowed.

12 OPERATION UNDER OVERLOAD CONDITIONS

This clause of IS 302-1 (1979) is applicable except as follows :

12.1 Addition

The test of **12.2** is not carried out if supplied from an incorporated transformers.

13 ELECTRICAL INSULATION AT OPERATING TEMPERATURE

This clause of IS 302-1(1979) is applicable except as follows :

13.1 Addition

Soldering guns and other tools supplied from a transformer are operated at a voltage equal to 1.06 times the rated voltage.

14 RADIO AND TELEVISION INTERFERENCE SUPPRESSION

This clause of IS 302-1 (1979) is applicable.

15 MOISTURE RESISTANCE

This clause of IS 302-1 (1979) is applicable except as follows :

Additional sub-clause

15.101 Household film-welding appliances provided with a suction device shall be constructed so that suction of liquid does not impair the electrical insulation.

NOTE – This requirement does not apply to parts operating at safety extra low voltage.

Compliance is checked by the following test:

40 ml of water containing approximately 1 percent NaCl is poured into a plastic bag. The bag is placed so that the surface of the water is at the same level as the

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suction opening of the appliance. The bag is then welded with the suction device in operation.

The appliance shall then withstand an electric strength test as specified in 16.4 and inspection shall show that water which may have entered the appliance does not impair compliance with this standard. In particular, there shall be no trace of water on insulation which could result in a reduction of creepage distances and clearances below the values specified in 29.1.

16 INSULATION RESISTANCE AND ELECTRIC STRENGTH (AFTER HUMIDITY TREATMENT)

This clause of IS 302-1 (1979) is applicable.

17 OVERLOAD PROTECTION

This clause of IS 302-1 (1979) is applicable except as follows:

17.1 Addition

The test is not carried out on soldering guns and other tools where the heating element is a part of the secondary circuit of the incorporated transformer.

18 ENDURANCE (Under consideration)

19 ABNORMAL OPERATION

This clause of IS 302-1 (1979) is applicable except as follows :

19.1 Addition

The test of 19.3 to 19.10 are not carried out on soldering iron and other tools where heating element is part of the secondary circuit of the incorporated transformer.

For heat guns and hot-air fire lighters compliance is also checked by the test of 19.101.

19.2 Replacement

Instead of operation without adequate heat discharge, tools are operated under the conditions specified in 11, except that soldering guns and other tools where the heating element is part of the secondary circuit of the incorporated transformer are operated continuously for 30 min, unless a biased-off switch is provided. In which case the tool is operated for 5 min.

19.2 Addition

Paint strippers incorporating integral scrapers are held horizontally in a clamp over the total length of the handle. A force of 70 N is exerted on the scraper edge in the direction corresponding to normal use.

Fire lighters are operated continuously for 2 h or until steady conditions are established, whichever is the longer; without adding more fuel.

19.5 Addition

Thermoplastic conduit-welding tools are operated with the fastest possible sequence of welding operations until steady conditions are established.

Additional Subclause

19.101 Heat guns and hot-air fire lighters are subjected to the following test, which is made with the heating element operated at the voltage specified in 11.4. The motor is supplied from a separate source and operated at its working voltage until steady conditions are established.

The motor speed is then reduced by decreasing the supply voltage until it is just sufficient to prevent the thermal cut-out from operating. The rate of decrease is:

- 1 V per min for motors having a working voltage not exceeding 30 V; and
- 5 V per min for motors having a working voltage exceeding 30 V.

The tools are then operated until steady conditions are established.

During the test the tools shall not emit flames or molten metal and the temperature rises shall not exceed the values specified in 19.11.

20 STABILITY AND MECHANICAL HAZARDS

This clause of IS 302-1 (1979) is applicable except as follows :

20.1 Addition

Hand-held tools shall have adequate stability when placed on their stand.

Contact fire lighters shall also have adequate stability when in use.

Compliance is checked by the following additional test:

The Fire lighter is placed on a horizontal surface such as a table so that the maximum insertion mark specified in 7.1 aligns with the edge of the surface. The power supply cord is freely suspended, its length including the plug not exceeding 1 m. The fire lighter shall remain stable in this position.

20.2 Addition

Soldering irons intended for de-soldering purposes, having a mechanism for operating an aspirating device, shall present no hazard to the user. In particular no spring loaded plunger which could strike the user may be incorporated, neither shall molten solder be accidentally ejected from the appliance when used in accordance with the manufacturer's instructions.

21 MECHANICAL STRENGTH

This clause of IS 302-1 (1979) is applicable except as follows:

21.1 Addition

For hand-held tools, compliance is also checked by the tests of 21.101.

For contact fire lighters, compliance is checked by the test of 21.102.

21.101 Additional Sub-clause

The power supply cord of hand-held tools is cut to a length of 10 cm, measured from the point where the cord, or cord guard, enters the tool.

The tool is then dropped from a height of 1 m on to a hardwood base having a thickness of 50 mm.

This test is made five times, the tool being in a position such that its major axis is horizontal and so that a different part of the tool is exposed to the impact each time.

The tool is then dropped five times, with its major axis vertical and with the soldering tip, on corresponding part of other tools, pointing downwards.

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After these tests, the tools shall show no damage within the meaning of this standard; in particular, live parts shall not have become accessible.

21.102 The handle of a new contact fire lighter is held firmly between two lightly padded surfaces with the heating element in a horizontal plane. The fire lighter is energized at rated voltage. After 3 min a mass of 4.5 kg is suspended at the end of the heating element for 1 min. The mass is removed and the element is allowed to cool. If the element has been bent, it is straightened to its original position.

After this test, live parts shall not be accessible and the fire lighter shall withstand the electric strength test of 16.4.

22 CONSTRUCTION

This clause of IS 302-1 (1979) is applicable except as follows :

22.1 Replacement

Soldering irons having a rated input not exceeding 200 W shall be Class I, Class II or of Class III.

Dehorning tools shall be of Class III having a rated voltage not exceeding 24 V.

Other tools shall be of Class I, Class II or Class III.

Compliance is checked by inspection and by the relevant test.

22.2 Addition

Transformers used for Class III dehorning tools shall be at least of splash proof construction. Thermoplastic conduit-welding tools shall be at least splash-proof.

Additional Sub-clause

22.101 Hand-held tools intended to be used away from the workshop shall have an integral stand.

Hand-held tools intended to be used on a table or similar surface shall have an incorporated or separate stand.

These requirements do not apply to tools withstanding the test of 11.

Examples of hand-held tools intended to be used away from the workshop are :

- Conduit-soldering tools;
- Thermoplastic conduit-welding tools;
- Branding tools;
- Dehorning tools;
- Heat-guns; and
- Paint strippers.

Examples of hand-held tools intended to be used on a table or similar surface are :

- Soldering irons;
- Desoldering irons;
- Burning-in pens; and
- Fire lighters.

22.102 Desoldering irons shall be provided with a device for collecting the solder.

Compliance with the requirements of **22.101** and **22.102** is checked by inspection.

22.103 Soldering guns shall be provided with biased 'OFF' switch which cannot be locked in 'ON' position.

Compliance is checked by inspection and by manual test.

22.104 Tools which are likely to suck liquid in normal use shall be provided with a drain hole at least 5 mm in diameter of an area of 20 mm² with a minimum width of 3 mm.

Compliance is checked by inspection and by measurement.

22.105 Thermoplastic conduit-welding tools shall be provided with a timer such that:

- It disconnects all poles of the welding circuits;
- After its cycle has been completed, the tool has to be reset before further operation; and
- It prevents a repetition of the welding operation on the same workpiece.

Compliance is checked by inspection.

NOTE – Repetition of the welding operation after disconnection of the input circuit is ignored.

22.106 The thermoplastic material surrounding the heating wire of fittings of thermoplastic conduit-welding tools shall ensure at least basic insulation when delivered.

Compliance is checked by the relevant tests.

23 INTERNAL WIRINGS

This clause of IS 302-1 (1979) is applicable.

24 COMPONENTS

This clause of IS 302-1 (1979) is applicable except as follows :

24.1 Addition

Suitably incorporated in tools which are not intended exclusively for household use shall be switches for frequent operation.

25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CABLES AND CORDS

This clause of IS 302-1 (1979) is applicable except as follows :

25.4 Addition

Type 2 attachment is allowed for :

Class III tools; and

Other tools, unless they are provided with a polyvinyl chloride insulated power supply cord and the temperature rise of external metal parts exceeds 75°C.

25.6 Replacement

For Class III tools and other hand held tools, the power supply cord shall not be lighter than :

If polyvinyl chloride insulated, light polyvinyl chloride sheathed flexible cords as per IS 694 : 1990.

If rubber insulated ordinary tough rubber sheathed flexible cord as per IS 9968 (Part 1) : 1988.

PVC cords are not permitted for thermoplastic con-

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duit-welding tools and fire lighters.

The length of the power supply cord for fire lighters shall be at least 1.50 m.

The length of the power supply cord shall be at least 4.0 m for Class III dehorning tools and at least 6.0 m for Class II dehorning tools.

For tools provided with a biased-off switch and for hand-held tools having a rated input not exceeding 100 W and a mass, without the power supply cord, not exceeding 100 g, polyvinyl chloride insulated power supply cords are permitted, regardless of the temperature rise of external metal parts.

25.10 Para Replacement

Instead of the number of flexings specified, the following applies :

20 000 for tools with a type 2 attachment; and
10 000 for other tools.

25.11 Para 15 Replacement

Instead of the first line in the table, the following applies for hand-held tools :

<i>Mass of the Tool</i>	<i>Pull</i>	<i>Torque</i>
kg	n	Nm
Up to and including 0.3	15	0.05
Over 0.3 up to and including 1	30	0.01

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of IS : 302-1 (1979) is applicable except as follows :

26.7 Addition

Rated current of appliance	Minimum nominal thread diameter	Minimum diameter of hole for conductor	Minimum length of thread in pillar	Maximum different between diameter of hole and nominal thread diameter
A	mm	mm	mm	mm
Up to and including 3	2.5	2.5	1.8	0.5
Over 3 up to and including 6	2.8	2.8	1.9	0.5

26.8 Addition

Rated current of appliance	Nominal thread diameter	Length of thread on screw	Length of thread in screw hole or nut	Nominal difference between diameter of head and shank of screw	Height of head of screw
A	mm	mm	mm	mm	mm
Up to and including 3	2.5	3.0	1.2	2.5	1.4
Over 3 up to and including 6	2.5	3.5	1.2	3.0	1.8

26.9 Addition

Rated current of appliance	Nominal thread diameter (minimum)	Difference between thread diameter and	
		inner diameter of washers (maximum)	Outer diameter of washers (minimum)
A	mm	mm	mm
Up to and including 3	2.5	0.3	3.0
Over 3 up to and including 6	3 ⁽¹⁾	0.4	3.5

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27 PROVISION FOR EARTHING

This clause of IS 302-1 (1979) is applicable except as follows :

27.1 Addition

Soldering irons of clause 11 intended to be used for soldering electronic equipment may be provided with equipotential bonding terminals without dimensional requirements.

28 SCREWS AND CONNECTIONS

This clause of IS 302-1 (1979) is applicable.

29 CREEPAGE DISTANCES AND CLEARANCES

This clause of IS 302-1 (1979) is applicable.

30 RESISTANCE TO HEAT, FIRE AND TRACKING

This clause of IS 302-1 (1979) is applicable.

31 RADIATION HAZARDS

This clause of IS 302-1 (1979) is not applicable.

101 TESTS

101.1 Type Tests

The tests specified in Table 101 shall constitute the type tests and shall be carried out on two samples of the same type and rating selected preferably at random from a regular production lot. Before commencement of the tests, the soldering irons shall be visually examined and inspected for obvious visual defects in respect of components, parts and their assembly, construction, mechanical hazards, markings, provision of suitable terminals for supply connections, earthing and the effectiveness of screws and connections. The external surface finish shall be even and free from finishing defects.

101.1.1 Criteria of Acceptance

Both samples shall successfully pass all the type tests for providing conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority at its discretion, may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test(s) in which failure(s) had occurred. No failure should be permitted in the repeat test(s).

Table 101 Schedule of Type Tests
(Clause 101.1)

Sl No.	Clause Reference
i) Protection against electric shock	8
ii) Starting of motor operated appliances	9
iii) Input	10
iv) Temperature rise	11
v) Operation under overload conditions	12
vi) Electrical insulation and leakage current at operating temperature	13
vii) Radio and television interforce suppression	14
viii) Moisture resistance	15
ix) Insulation resistance and electric strength (after humidity treatment)	16
x) Overload protection	17
xi) Endurance (Under consideration)	18
xii) Abnormal operation	19
xiii) Stability and mechanical hazards	20
xiv) Mechanical strength	21
xv) Construction	22
xvi) Internal wiring	23
xvii) Components	24
xix) Cord grip and cord guard	25
xx) Terminals for external conductors	26
xxi) Provision for earthing	27
xxii) Screws and connections	28
xxiii) Creepage distance and clearances	29
xxiv) Resistance to heating and tracking	30
xxv) Resistance to rusting	31

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101.2 Acceptance Tests

The following shall constitute the acceptance tests:

<i>Tests</i>	<i>Clause Reference</i>
a) Protection against electric shock	8
b) Input	10
c) Temperature rise	11
d) Electrical insulation and leakage current at operating temperature	13
e) Moisture resistance	15
f) Insulation resistance and electric strength (after humidity treatment)	16
g) Provision for earthing	27

101.2.1 A recommended sampling procedure for acceptance tests is giving in Appendix B of IS 302-1 (1979).

101.3 Routine Tests

The following shall constitute the routine tests :

<i>Tests</i>	<i>Clause Reference</i>
a) Protection against electric shock	8
b) High voltage	13.3.2 of IS 302-1 (1979)
c) Provision for Earthing	27

ANNEX A

TABLE OF TYPE TESTS

The Appendix A of IS 302-1 (1979) is not applicable.

ANNEX B

SAMPLING BROACHER FOR ACCEPTANCE TESTS

The Appendix B of IS 302-1 (1979) is applicable.

ANNEX C

ELECTRONIC CIRCUITS

The Appendix C of IS 302-1 (1979) is applicable if electric circuits are used.

ANNEX D

MEASUREMENT OF TEMPERATURE WITH THERMOMETER

The Appendix D of IS 302-1 (1979) is applicable.

ANNEX E

ALTERNATIVE TEST FOR PROTECTED MOTOR UNITS

The Appendix E of IS 302-1 (1979) is applicable.

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ANNEX F

IMPACT TEST ALTERATION

The Appendix F of IS 302-1 (1979) is applicable.

ANNEX G

THERMAL CONTROLS AND OVERLOAD RELEASER

The Appendix G of IS 302-1 (1979) is applicable.

ANNEX H

MEASUREMENT OF CREEPAGE DISTANCE AND CLEARANCES

The Appendix H of IS 302-1 (1979) is applicable.

ANNEX J

TEST FOR FIRE RESISTANCE PROPERTIES

The Appendix J of IS 302-1 (1979) is applicable.

ANNEX K

BNF JET TEST FOR DETERMINATION OF THICKNESS OF COPPER AND NICKEL PLATING

The Appendix K of IS 302-1 (1979) is not applicable.

ANNEX L

APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM ON NICKEL, STEEL AND COPPER

The Appendix L of IS 302-1 (1979) is not applicable.

ANNEX AA

(Foreword)

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
302-1 (1979)	Safety of household and similar electrical appliances : Part 1 General requirements (<i>fifth revision</i>)	1416 : 1972	Safety transformers (<i>first revision</i>)
694 : 1990	PVC insulated cable for working voltages up to and including 1 100 V (<i>third revision</i>)	9968 (Part 1) : 1988	Elastomer insulated cables : Part 1 For voltage up to and including 1 100 V (<i>first revision</i>)

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